

HackMe Report

GITHUB REPO: <https://github.com/shadielfares/Bravo.git> (see README.md to run program)

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Problem Statement:

In 2020 Version conducted a data breach investigations report, and analysed over 40,000 security incidents where they found that over a quarter of the attacks targeted small businesses. IBM also found that a data breach on average costs companies \$3.9 million! HackMe helps these small businesses secure their data by scanning for vulnerabilities using a variety of penetration methods. Designed for ease of use, it identifies potential weaknesses, enabling companies to assess and enhance their website security without requiring extensive technical expertise, and presents a well-formatted report that suggests methods of improving cybersecurity standards.

Summary of Solution's Objectives:

This project aims to help startups secure their websites by automating the detection of common vulnerabilities. The first objective is to implement an automated port scanning feature using the `nmap` library. This feature scans a target website or server for open ports that may present security risks. By checking a specified target and port (e.g., port 3000), the port scanner returns structured data, such as a list of open ports that could be exploited by attackers. This helps identify potential entry points that could compromise the system.

The second objective is to test for weak login credentials using brute-force attacks. The project utilizes the `paramiko` library to attempt logins on a server via SSH or HTTP using predefined username and password combinations. If the login attempt is successful, it indicates weak credentials that should be updated. This process helps detect vulnerabilities caused by weak login configurations that attackers can easily exploit. The system will report the outcome of each login attempt, specifying which combinations were successful and which failed.

The final objective is to create a user-friendly interface that makes the tool accessible for startups with limited cybersecurity expertise. This interface will allow users to initiate scans and view the results through a web dashboard. By automating vulnerability detection and presenting the findings in an easy-to-understand format, this project empowers startups to proactively address security risks without needing in-depth technical knowledge. Overall, the goal is to provide startups with an easy-to-use security tool that automates vulnerability checks and helps them safeguard their online presence by identifying and addressing potential cyber threats.

Summary of Contributions:

1. Backend Development and Vulnerability Automation

- **Responsibilities of Team Member 1:** Team Member 1 was responsible for setting up the backend environment and automating vulnerability scans. The initial task involved establishing the Python project by setting up a virtual environment and installing essential libraries like nmap, python-nmap, and paramiko. A port scanner was implemented using nmap to scan websites for open ports and identify potential vulnerabilities. Additionally, a basic brute-force login tester for SSH or HTTP was developed to attempt login with a small set of username and password combinations. The structured data was then prepared for display on the frontend.

2. GPT Integration and Report Generation

- **Responsibilities of Team Member 2:** Team Member 2 integrated the GPT API to create professional reports from the structured vulnerability data received from the backend. These reports were designed to present clear, client-friendly information, including the severity of vulnerabilities, potential risks, and actionable recommendations. A customizable report template was developed to tailor output based on the specific vulnerabilities identified during the scan. After the integration, dummy data was used to test and validate the report generation, ensuring accuracy and user-friendly presentation.

3. User Interface and System Design

- **Responsibilities of Team Member 3:** Team Member 3 focused on developing the user interface and the server.py file. By using a React front end and a flask back end. Team member 3 was responsible for the architecture of the project. A clean web page was designed that takes a hyperlink that would be scraped for vulnerabilities, and this was sent to the API endpoint that Team Member 3 designed. A post request was used to send the link, and the pdf was returned which is automatically downloaded by the client.

4. Coordination, Documentation, and Local Deployment

- **Responsibilities of Team Member 4:** Team Member 4 ensured all the components worked together and locally hosted the project throughout development. Additionally, they ensured the project was well documented by writing the README.md file to ensure clear instructions and project details for future contributors. The project was tested and debugged with the help of the other team members to confirm that all functions operate as desired. A demo was prepared by team member 4, highlighting the tool's ability to scan for vulnerabilities, generate comprehensive reports, and integrate seamlessly into security practices for startups and other business ventures.